

Greening The Exterior Walls



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Greening the exterior wall systems of residential structures is a complicated process, and it hasn't been made any easier by the emergence of alternative wall systems. Of course, these new systems, such as steel framing, structural insulated panels (SIPs), combination foam/steel panels, insulated concrete forms (ICFs), autoclaved aerated concrete walls (AACs), and strawbale walls ultimately have the potential to make a building more efficient, green, and attractive. Add to this the fact that typical wood-framed walls have been adapted in a number of ways to improve their effectiveness, and you have a lot of new information to absorb. Learning about all of these changes, and the best exterior system to use for each job, is a potentially daunting task. However, a little explanation can go a long way toward understanding what will be the best wall for the job. Of course, cost, no matter how effective the system, may be the determining factor.

Meeting Criteria

From a building performance perspective, any wall system must meet three basic criteria in addition to structural integrity. First, the wall system must include some effective flashings, and some form of rain screen that can keep water out of a wall, but be permeable enough to let the wall dry to the exterior or interior if it should get wet. This means that it should have no vapor barrier. With the increased use of air conditioning in heating climates, I have too often found that either a vapor barrier or vinyl wallpaper has trapped moisture in the walls, and mold is growing in the

wall cavity or in the drywall behind the wallpaper. Second, the wall system must provide an air barrier to keep air (especially moisture-laden air) from moving into and through the wall. It has been shown that air sealing exterior walls dramatically increases the energy efficiency of our homes, and dramatically reduces the amount of moisture entering our walls. And third, the wall system must provide an R-value appropriate to the heating and cooling zone in which it is built. (A good source for appropriate R-values can be found at www.energy.gov/ia/new_homes/features/IncreasedInsulation1-17-01.pdf.)

Advocates of green building would fully adhere to these three criteria. Changes that improve the durability and energy efficiency of a structure are essential to green or sustainable building. Green building would then add three criteria of its own to the deliberations as to which wall system would be best for the job.

First, the wall system must use the minimum of materials necessary to do its work without compromising the integrity of the structure. A good example of this is the improvements that have been made in wood framing. Optimum Value Engineering (OVE) means designing our homes so that they use common lengths of building materials that come in 2-ft increments. Also, if the architectural drawings show that each stud rests just above a joist and that the next joist rests directly above that stud, less top plate material is needed. Using framing that is 2 ft on center also reduces the amount of wood used. For years, wood was delivered to the work site uncut to specification, which meant that there was a lot of waste wood at the end of a project. Now, however, wood can be delivered in pre-cut specified lengths to a site; and the use of oriented strand board (OSB), finger-jointed studs, and so on means that wastewood can be used in place of bigger trees.

Second, the wall system must be constructed of renewable materials or materials that can be salvaged, recycled,

or biodegraded at the end of their useful life. Wood framing in good condition can be reused and waste wood is biodegradable. Steel framing can be recycled, and to do so uses as little as 15% of the energy that was used to manufacture it in the first place. Concrete can be recycled and used in a variety of ways in future construction sites.

And finally, the system should be constructed of materials that have the least impact on the environment from the time they are produced to the time they end their useful life. This requirement is the most difficult to achieve with the materials currently available.

Comparing Wall Systems

It would be best if all wood used could come from certified forests. There are not yet enough of these forests to come close to providing all the wood required, but we should at least search for certified sources in our area. I was able to locate a certified hardwood forest within 200 miles of my area that produces wood for a variety of oak products, like hardwood flooring.

Producing the cement for concrete to be used in an ICF wall requires enormous amounts of energy, and it releases a great deal of CO₂ into the earth's ozone layer. But the stone, gravel, and sand used in the concrete can usually be obtained locally, which greatly reduces shipping costs. Cement companies now use millions of old tires each year to fire their production furnaces at temperatures so high that no pollutant gases are released. One recent study demonstrates ways to incorporate the CO₂ emissions into the cement, which could prove to be the most effective solution.

Straw, a by-product of wheat harvesting that is used to construct strawbale homes, comes closest to being the perfect exterior wall material. It is easily obtained and easily prepared, and because it is locally accessible, the shipping costs are low.

Foams, used in many wall systems, have an environmental downside if they are produced using chlorofluorocarbon

(CFC) or hydrochlorofluorocarbon (HCFC) blowing agents. These can damage the ozone layer. But foams can be recycled, which is an up. The problem is that there is as yet no cost-effective way to get the waste foam back to the manufacturer for recycling. I took foam left over from sheathing work on our first homes to Mailboxes stores to use for shipping. This is a short-term solution that is really not a solution, because the foam is still likely to end up in a landfill one day.

Fiberglass insulation manufacturers now use up to 40% of recycled glass in the production of their products, but recycling fiberglass insulation after its use in a home is, at least for the time being, a dead end. While some people suggest reusing it, the dust, molds, and other pollutants that can infiltrate the fibers makes this a less than satisfactory solution. On the other hand, the elimination of formaldehyde in its manufacture is a plus.

Getting to Green

The first green homes I was associated with used the *Builder Guide for Cold Climates* to guide the framing. We used 2 x 6 wood framing, 24 inches on center, with foam sheathing on the exterior and R-19 insulation batts on the interior. The exterior sheathing and the interior drywall were sealed to the framing; all seams were sealed and openings effectively flashed. The framing drawings aligned stud over floor joist and floor joist over stud, so only single top plates were required. We met most of the six criteria, while recognizing that most of the products we used had their limitations. Just raising the issues is a significant first step, and finding the best materials and systems you can, given what is currently available, is a good second step. Providing a wall system that meets all six criteria requires a lot of attention to detail and continuous monitoring during construction. But to date, setting the criteria has proved to be very effective.

What about steel-framed walls with foam sheathing? The foam sheathing is a must to avoid heat loss through this framing, since the R-value of steel is much less than that of wood. Air sealing a steel-framed wall will require the same attention to detail as air sealing a wood-

framed wall. In addition, using steel means using a nonrenewable resource, though steel is fully recyclable. The foam/steel walls meet most of the other criteria, except that the foam cannot currently be recycled.

SIPs also meet most of the home performance and green building criteria. These panels also make air sealing and flashing much easier to accomplish



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if one follows manufacturer specifications. Air sealing points of connection between the wall and roof panels, and points where the SIPs interface with other building components (such as floors) is all that is usually necessary, since the SIPs themselves are sealed in the manufacturing process. Of course, an effective drainage plane is still necessary on the exterior. The panels contain foam, with its limitations regarding HCFC and recycling. But manufacturers of SIPs are required to meet EPA regulations for the use of HCFC, and a number of alternatives are emerging that should greatly reduce this problem.

ICFs also meet all of the home performance and green building criteria, excluding the foam issue. While cement and concrete are not renewable resources, the concrete can be fully

recycled and used in a variety of ways. Installing ICFs correctly addresses the air sealing challenge.

AAC walls, a recent entry in the building materials/system market, meet all the criteria except that of being derived from a renewable resource; but AAC is fully recyclable at the end of its useful life. What makes AAC walls particularly interesting to those of us concerned with building low-to moderate-income housing is that they do not require the addition of any siding, insulation, or drywall. This could dramatically reduce the cost of construction.

Without question, strawbale construction most fully meets all six criteria, and like AAC walls, it requires no siding, insulation, or drywall. If it were not so labor intensive to construct, strawbale would be the wall system of choice. I recently spent some time helping a woman in our area construct a strawbale home, guided by a strawbale expert. The most critical issue when using this system is that the bales cannot get wet during construction. If they do, they must be replaced. However, once they are in place and the surface clay/straw coatings are installed on the inside and outside, wetting is not a problem, because the surfaces allow the walls to dry effectively.

While each of the wall systems mentioned in this column has upsides and downsides, both from a home performance and from a green building perspective, some systems do meet most, if not all, of our criteria. Foam is widely used in exterior wall systems to produce the insulation values we want in the homes we build. However, from a green building perspective, foam has a downside as well. If a more environmentally friendly version of foam can be produced; if recycling efforts can be ramped up over the next few years so that non-renewable building materials can be effectively recovered; and if our designs for residential construction can make use of less material while still producing a quality product, our buildings will be performing in a way that benefits their residents and the environment from which their components were taken.

